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ConsRAG: Minimize LLM Hallucinations in the Legal Domain

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Abstract. Retrieval-Augmented Generation (RAG) systems have shown potential in improving legal question-answering applications. However, they often struggle to provide precise information for legal queries, as broad-topic relevance may not always align with contextual usefulness. To address this challenge, we introduce Constrained Retrieval-Augmented Generation (ConsRAG), a novel approach that employs aspect-based constraints during both retrieval and generation phases. ConsRAG aims to enhance precision and contextual relevance in legal outputs through these constraints and an iterative backtracking mechanism. Our experiments suggest that ConsRAG may offer improvements over existing systems in terms of accuracy and relevance of retrieved documents. This paper presents the framework, implementation, and evaluation of ConsRAG, exploring its potential to enhance the reliability of legal AI applications.

Keywords. Constraint-based Generation, Hallucinations Mitigation, ConsRAG

1. Introduction

Precise legal question-answering [1,2,3,4] is crucial in the legal field, where accurate information significantly influences decisions. Traditional systems often struggle with retrieving relevant documents [5], while generative models risk producing hallucinations [6,7]. Retrieval-Augmented Generation (RAG) [8,9,10] offers a hybrid solution, but its efficacy in legal QA is often hindered by irrelevant document retrieval. Current RAG implementations (see [11,12,13]) often fail to incorporate aspect-based relevancy, leading to issues with retrieving wrong documents and generating inaccurate responses. Simply put, a retrieved document may be relevant, but if it doesn't contain information about the aspect that helps answer the question, it has no value. This is particularly problematic in the legal domain, where precision and contextual understanding are critical.

To address these challenges, we propose Constrained Retrieval-Augmented Generation (ConsRAG), a novel approach for legal QA. ConsRAG introduces conditional constraints during retrieval and generation processes to ensure precision and contextual relevance, aiming to mitigate hallucinations and improve accuracy. The ConsRAG framework brings significant advancements to the field of legal question-answering. It integrates aspect-based constraints during both the retrieval and generation phases, ensuring that the information gathered is both relevant and precise. Additionally, ConsRAG implements an iterative backtracking mechanism, which enhances the accuracy and re-

Aspect	**Description**
Numerical Values	Legal texts often involve specific numerical values, such as ages, monetary amounts, durations, fines, and penalties. Accurate extraction and use of these values are essential for providing precise legal answers.
Legal Definitions	The use of specific legal terms and their precise definitions ensures clarity and legal accuracy in responses. These definitions must be consistently retrieved and applied.
Case Citations	References to relevant legal cases are fundamental in legal reasoning. Accurate citation of these cases helps ground generated responses in established legal precedents.
Statutes and Regulations	Specific laws and regulations cited in legal queries must be accurately identified and applied to ensure responses are compliant with the relevant legal frameworks.
Entity Recognition	Proper identification of entities such as individuals, corporations, and organizations is crucial for contextual relevance and accuracy in legal QA.
Dates and Timelines	Legal queries often involve specific dates and chronological events. Correctly integrating these timelines is crucial for coherent and accurate answers.
Outcomes and Penalties	Information regarding potential legal outcomes and penalties must be precisely retrieved and generated to address the core of many legal inquiries.
Jurisdictional References	Accurately incorporating details about specific jurisdictions ensures that responses are legally sound and contextually appropriate.
Procedural Requirements	Many legal questions involve procedural aspects. Reflecting accurate procedural requirements ensures compliance with legal processes.
Legal Principles	References to general legal doctrines or principles provide robustness and coherence to legal arguments and responses.

Table 1. Common Aspects in Legal QA and Considered in Our Framework

liability of the answers provided. The framework offers flexibility in selecting aspects, allowing it to adapt to different legal contexts and queries. By systematically addressing the limitations found in traditional RAG systems, ConsRAG provides a robust solution for incorporating AI into legal domains. This approach is crucial for meeting the high standards of precision and accuracy required in legal question-answering systems.

2. ConsRAG Framework

Aspect-based analysis is the foundation of the ConsRAG framework, guiding the retrieval and generation processes. This involves identifying and examining key components of the QA task, such as document relevance, contextual coherence, legal accuracy, and information completeness. Through aspect analysis, we pinpoint specific details crucial for legal question-answering task.

The aspects in Table 1 serve as guiding parameters in our system.

In the retrieval phase, documents are scored for relevance based on aspect-specific criteria. The relevance score $R(d_i, Q)$ for each document d_i with respect to the query Q is calculated using a relevance function $f(d_i, Q)$:

$$R(d_i, Q) = f(d_i, Q)$$

This scoring mechanism ensures that retrieved documents meet the required aspect-based relevance criteria, filtering out non-pertinent texts. For example, a query inquiring about the legal age for marriage would prioritize documents with numerical age information over those discussing general marital topics.

During the generation phase, the generated output is validated against the retrieved documents. Consistency checks ensure that the generated response incorporates all critical information relevant to the query's aspects. The consistency $C(G,D,Q)$ of a generated answer G with respect to documents D and query Q is calculated as follows:

$$C(G,D,Q) = \sum_{i=1}^{n} (\text{similarity}(G,d_i) \cdot R(d_i,Q)) \cdot R(G,Q)$$

Where $R(G,Q)$ is the relevance of the generated response G with respect to the query Q, ensuring that the generated response is relevant to the query's aspects, such as numerical values if the query involves specific numerical information. This mechanism aligns the generative model's output with aspect-based constraints, minimizing hallucinations.

The ConsRAG framework includes an iterative backtracking mechanism that ensures quality control throughout the retrieval and generation stages. If constraints are not met, the system re-evaluates and refines the steps iteratively:

1. **Initial Retrieval Phase**: Retrieve documents based on query Q.
2. **Relevance Scoring**: Calculate relevance $R(d_i,Q)$ for each document d_i.

$$R(d_i,Q) = f(d_i,Q) \tag{1}$$

$$\text{Relevance Threshold Check: } R \geq R_{th} \tag{2}$$

3. **Generation Phase**: Generate an initial answer G based on documents D.
4. **Consistency Scoring**: Calculate consistency $C(G,D,Q)$.

$$C(G,D,Q) = \sum_{i=1}^{n} (\text{similarity}(G,d_i) \cdot R(d_i,Q)) \cdot R(G,Q) \tag{3}$$

$$\text{Consistency Threshold Check: } C \geq C_{th} \tag{4}$$

The backtracking mechanism in the ConsRAG framework operates by first adjusting the generation phase. If the consistency score C falls below the threshold C_{th}, the model is prompted to regenerate the response. Should repeated attempts at regeneration continue to fail, the system then backtracks to the retrieval phase to refine the query and retrieve more relevant documents. This process is demonstrated in Algorithm 1. This iterative process ensures that both retrieval and generation phases meet the necessary constraints for accuracy and relevancy, thereby minimizing hallucinations and enhancing the overall reliability of the system.

Algorithm 1. ConsRAG Iterative Backtracking Mechanism

Require: Query Q, Thresholds R_{th} and C_{th}

Initialize $D \leftarrow \text{RetrieveDocuments}(Q)$
repeat
    $R \leftarrow \text{CalculateRelevance}(D, Q)$
    if $R < R_{th}$ **then**
        $Q \leftarrow \text{RefineQuery}(Q)$
        $D \leftarrow \text{RetrieveDocuments}(Q)$
    end if
until $R \geq R_{th}$
repeat
    Initialize $G \leftarrow \text{GenerateAnswer}(D)$
    $C \leftarrow \text{CalculateConsistency}(G, D, Q)$
    if $C < C_{th}$ **then**
        $G \leftarrow \text{RegenerateAnswer}(D)$
        if C still $< C_{th}$ **then**
            Backtrack to refine D: $D \leftarrow \text{RetrieveDocuments}(Q)$
        end if
    end if
until $C \geq C_{th}$
return G

3. Experiments

To evaluate the performance of the ConsRAG framework, we utilize the COLIEE (Competition on Legal Information Extraction/Entailment) dataset [14]. COLIEE is an annual competition that focuses on several tasks related to legal information processing. For our experiments, we use Task 3 to evaluate the retrieval component of ConsRAG, and data from Task 4 to formulate constraints for the generation component. We allow the LLM to autonomously decide the aspects it will use to assess the relevance of articles and the expected response after analyzing the query.

While we leverage both Task 3 (Statute Law Retrieval) and Task 4 (Legal Textual Entailment) from the COLIEE dataset in our experiments, our primary evaluation focuses on Task 3, as our main aim is to improve retrieval performance. The performance is assessed using the macro average of F2, precision, and recall. Precision measures the proportion of correctly retrieved articles among the retrieved ones, while recall measures the proportion of correctly retrieved articles among all relevant articles. The F2 score gives more weight to recall, highlighting the importance of not missing relevant documents in legal contexts.

The results of the experiment are visualized in Figure 1, which compares the F2 scores of the ConsRAG framework with top-performing models from the COLIEE 2024 competition [14]. The ConsRAG framework exhibited a promising performance compared to other baseline models, with the iterative backtracking mechanism linking the retrieval and generation phases contributing to notable improvements in retrieval precision and the overall F2 score. The framework achieved an F2 score of 0.83, which was the highest among the tested models. Several key observations merit attention: firstly, the application of aspect-based constraints during both the retrieval and generation phases ap-

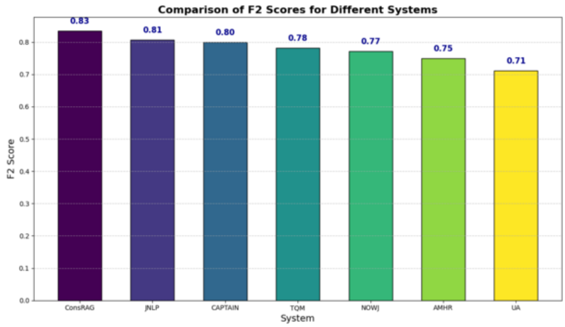

Figure 1. Comparison of F2 Scores for Different Systems. See [14] for detailed descriptions of the compared systems.

peared to enhance relevance and accuracy. Secondly, the dynamic backtracking mechanism, which allowed the system to revisit the retrieval phase when generation constraints were not satisfied, seemed to support improved performance and a reduction in errors and hallucinations. Lastly, consistency checks during the generation phase appeared to significantly reduce hallucinations, thereby aiding in grounding generated responses within the retrieved legal documents.

4. Conclusions

In this paper, we introduced the Constrained Retrieval-Augmented Generation (ConsRAG) framework to enhance the accuracy and reliability of legal question-answering systems. By integrating aspect-based constraints during both the retrieval and generation phases, and employing an iterative backtracking mechanism, ConsRAG minimizes hallucinations and ensures contextual relevance. Evaluated on the COLIEE 2024 dataset, ConsRAG outperformed state-of-the-art systems, demonstrating its effectiveness in legal QA tasks. Our framework's flexibility allows for user-defined or autonomously generated aspects, making it adaptable to various legal contexts. Future work will focus on enhancing retrieval alignment with gold standards, optimizing computational efficiency, expanding applicability to multilingual datasets, and improving explainability to further solidify ConsRAG's role in legal AI applications.

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